Thinking vs. Doing: Agents that Reason by Scaling Test-Time Interaction

Junhong Shen, Hao Bai, Lunjun Zhang, Yifei Zhou, Amrith Setlur, Shengbang Tong, Diego Caples, Nan Jiang, Tong Zhang, Ameet Talwalkar, Aviral Kumar

Paper version: arXiv 2506.07976v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *This process does not allow agents to acquire new information from the environment.*

TTI scales the **number of interaction steps**, not per-step thinking: explore, backtrack, re-plan in one rollout.

Motivation: a new axis of test-time scaling

1. **Thinking more** reorganizes known info.
2. **Acting more** gains new info from the environment.
3. **Prompted re-checks** confuse; training must steer.

Evaluation setup

WebVoyager · 427 | WebArena · 812 | Gemma 3 12B base | 27B judge | h: 10 → 30 | online eval

- **Online** task success, original evaluators; the judge trains only.
- **Data:** 128K (WV) / 11K (WA) self-made synthetic tasks.

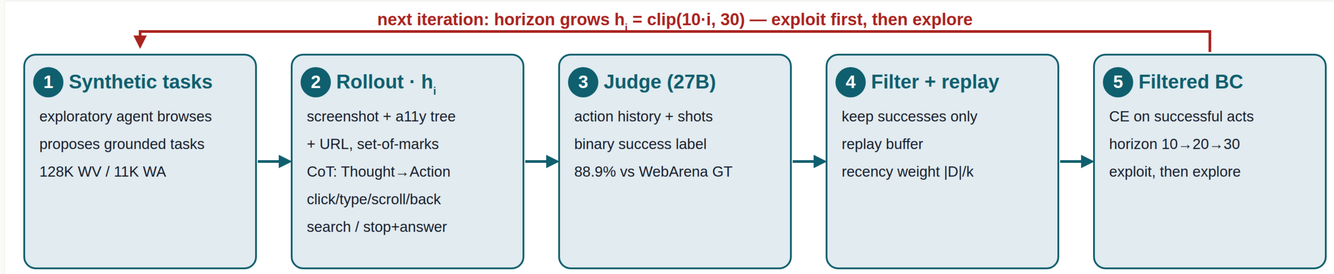
Our read (CUA Radar)

61	★★	online
score of 100	band	evidence level

- **Reports:** online eval; same-base fixed-horizon baselines; token-budget compare; cost; curriculum ablation; judge 88.9%; code.
- **Not reported:** seeds/variance for main tables; held-out env partial. **Flags:** none.
- Pattern K20. Authors' numbers, not rerun.

Method: TTI — filtered BC under a rollout-horizon curriculum

- **Sees / acts:** screenshot + a11y tree + URL in; click, type, scroll, go back, search, stop out; history keeps all actions, last 3 observations.
- **Curriculum:** multiplicative horizon $h_i = \text{clip}(h_{min} \cdot i, h_{max})$: 10 → 30, with recency-weighted replay of successes.
- **Signal:** binary success from a Gemma 3 27B judge; online filtered BC trains only on successful on-policy rollouts.

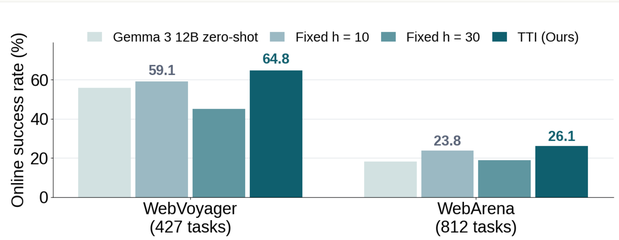


Our redraw (not the paper's figure): the TTI loop — synthetic tasks, horizon-scheduled rollouts, judged, filtered, distilled back.

$$J(\theta) = \mathbb{E}\Big[\Big(\sum_{t=0}^{h-1} \log \pi_\theta(a_t \mid o_{\le t}, \mathcal{T})\Big) \cdot \mathbf{1}[R = 1]\Big]$$

Online filtered BC (Eq. 5.1): likelihood of successful rollouts only

Key results: online task success (%)

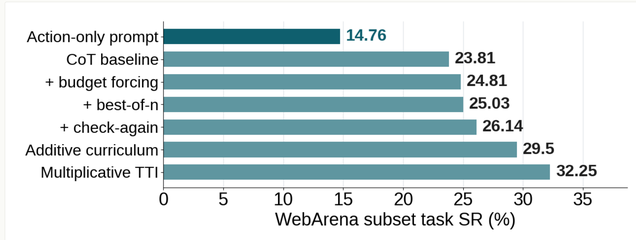


Tables 3–4, redrawn: all four series share the Gemma 3 12B base.

As reported: +9% WebVoyager, +8% WebArena over the zero-shot base

SO WHAT → *TTI is best on 8/13 WebVoyager domains; per-step CoT gets shorter as rollouts get longer.*

Same 62-task WebArena subset, 3 runs



Tables 1, 2, 6, redrawn: prompting strategies vs RL curricula; test limit h = 30.

SO WHAT → *Training the horizon (32.25) beats prompting it (check-again 26.14); multiplicative beats additive (29.50). Fixed h = 10 overfits early; fixed long horizons learn slowly. Re-checks on TTI checkpoints add modest gains, largest early in training.*

Headline Numbers

64.8%
WebVoyager success
zero-shot 55.8%

26.1%	WebArena success
88.9%	judge vs GT eval
8 / 13	best WV domains

The paper's own controls

CONTROL	RESULT	CONTEXT
Fixed h=10/30 (WV)	59.1 / 45.2	TTI 64.8
Fixed h=10/30 (WA)	23.8 / 19.0	TTI 26.1
Token budget	interaction best	BoN/forcing <3%
Curriculum	additive 29.50	TTI 32.25

Judge 88.9% vs GT eval; UI-TARS 84.8 needs private data

Takeaway

Scale steps, not thinking: a horizon curriculum over online filtered BC buys exploration and backtracking.

128K + 11K synth tasks | h 10 → 30 curriculum | 8/13 WV domains best

Scan to Read

Paper | Code